Arda Goktogan

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Research Profile

Computer Science M.S. student at Bilkent University researching 3D scene reconstruction and completion, neural rendering, and generative world models.

Education

Bilkent University Jan 2024 – Present
M.S. in Computer Science; expected graduation: Jan 2027
Research focus: 3D Computer Vision and Spatial Intelligence; Ankara, Turkey

Bilkent University Sep 2017 – Jun 2022
B.S. in Computer Science
GPA: 3.85/4.00; ranked 8/160; Ankara, Turkey

Publications

StyleFusion360: View-Consistent Head Stylization via Adaptive Style Modulation
Furkan Güzelant, **Arda Goktogan**, Tarık Kaya, and Aysegul Dundar.
European Conference on Computer Vision (ECCV), 2026. Paper | Code

Contribution: Studied shuffled prediction for cross-batch consistency in multiview diffusion and interpolation for combining styles; implemented masking in the proposed attention mechanism for masked style transfer.

Mukayese: Turkish NLP Strikes Back
Ali Safaya, Emirhan Kurtuluş, **Arda Goktogan**, and Deniz Yuret.
Findings of ACL, 2022, pp. 846–863. Paper

The Use of 3D Optical Flow, Feature-Tracking and Token-Tracking Petri Nets to Analyze and Visualize Multiple Scales of Ocean Eddies
Sedat Ozer, Karen Bemis, Weiping Hua, **Arda Goktogan**, Melike Aydoğan, Kevin Guo, Dujuan Kang, Li Liu, and Deborah Silver.
IEEE VIS, 2020 – SciVis Contest Paper. Honorable Mention. Contest record

Research Experience

Geometry-Aware 3D Scene Completion Sep 2025 – Present
M.S. Research Project, Bilkent University DLRLab
Advisors: Ayşegül Dündar and Duygu Ceylan

- Developed differentiable camera optimization to select views and regions for inpainting within world-model pipelines that generate scenes through image inpainting and video generation.
- Demonstrated a 2–3 point reduction in FID in experiments and prepared a research manuscript for submission.

Refedit: Zero-Shot Reference-Based Image Editing 2024 – 2025
Research Project, Bilkent University
With Ayşegül Dündar

- Developed zero-shot reference-based object insertion with automatic edit localization, without fine-tuning, paired training data, or a user-provided mask.
- Designed latent-space refinement for style, background preservation, and reference similarity; evaluated edit locality, fidelity, and visual coherence.

Earlier Research Experience

Koç University AI Lab Feb 2021 – Feb 2022
Student Researcher, Turkish Natural Language Processing; Remote
Advisor: Deniz Yuret

- Developed data preparation, evaluation, and training baselines for POS tagging, named entity recognition, and spell checking in the Mukayese benchmark, published in Findings of ACL 2022.
- Developed a Hunspell-based Turkish spell checker for LibreOffice. Code

Bilkent University OzerLab 2020 – May 2021
Student Researcher, Computer Vision and Robotics; Ankara, Turkey
Advisor: Sedat Ozer

- Implemented optical flow on volumetric ocean data for multiscale eddy analysis as part of the work awarded Honorable Mention at the IEEE SciVis Contest 2020.
- Evaluated Mask R-CNN for small-instance aerial segmentation; studied PRM, RRT, and RRT* and developed a similarity-domain-based robot path planner.

Technical Experience

ArgosAI Feb 2022 – May 2025
Computer Vision Engineer; Ankara, Turkey

- Developed transformer-based video classification and Mask R-CNN segmentation with a temporal backbone for airport runway foreign-object detection.
- Optimized real-time inference with TensorRT and implemented a multiprocessing Python pipeline for image and video processing.

Büyütech Summer 2021
Computer Vision Intern; Ankara, Turkey

- Evaluated lane-detection models on Turkish road imagery and ported the selected PyTorch model to C++/LibTorch on NVIDIA Jetson Xavier NX, achieving approximately 15× faster inference than the original Python implementation on the same device.

Technical Skills

Focus: 3D Gaussian Splatting, neural rendering, diffusion-based editing, computer vision.
Tools: PyTorch, OpenCV, TensorRT, LibTorch, TensorFlow, Keras, Linux.
Languages: Python, C++, C, Java, Julia, MATLAB; LaTeX.

Selected Honors and Awards

- Teknofest Turkish NLP Competition – **1st Place**, 2021.
- IEEE SciVis Contest – Honorable Mention, 2020.
- Ranked 240th among approximately 2 million participants in Turkey's Undergraduate Placement Examination, 2017.
- Bilkent University Comprehensive Scholarship, 2017–2022.